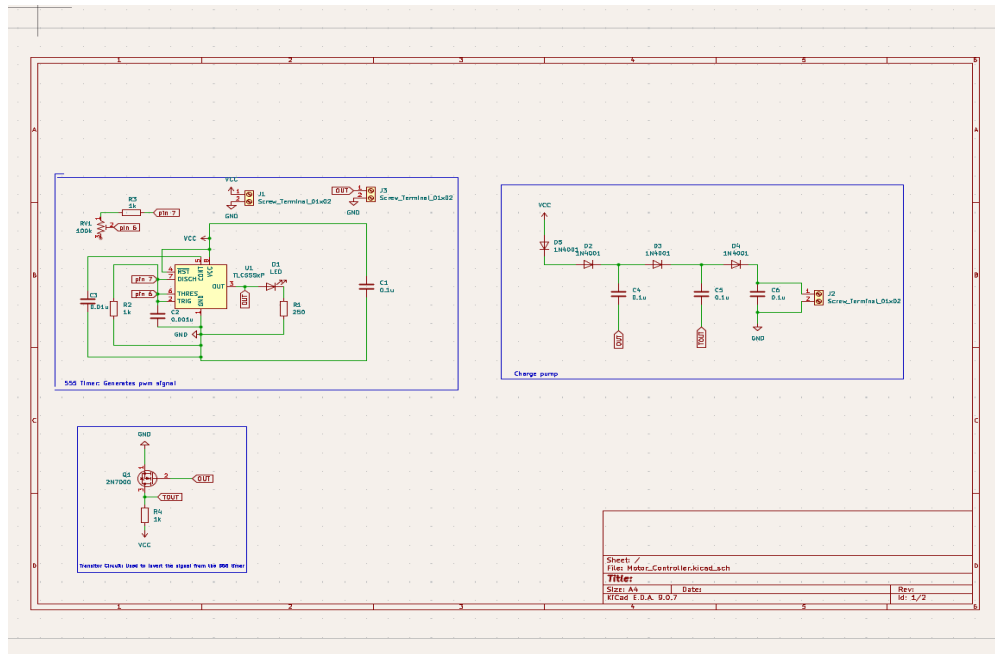


Summary

The objective of this project was to design and build a voltage multiplier using a 555 timer. The motivation for this project stemmed from my desire to develop a DC motor controller for a 9 V DC motor; however, I did not have access to a sufficiently high supply voltage. By using a 555 timer to generate an oscillating signal, the voltage multiplier circuit was able to produce a higher DC voltage from a lower-voltage source using components that were already available.

1) Schematic



1) 555 timer Oscillator

- Configured in astable mode using the TLC555 timer to generate a square-wave output signal.
- A 100 k Ω potentiometer (RV1) is used to adjust the oscillation frequency.
- Provides the switching signal required to drive the voltage multiplier circuit.

2) Charge pump to store and release charge

- Uses a 2N7000 MOSFET to generate an inverted version of the 555 timer output.
- Produces a complementary switching signal (IOOUT) from the original timer output (OUT).
- Ensures the voltage multiplier receives alternating charging signals for efficient charge transfer.

3) Transistor Circuit to invert the PWM signal due to multi-stage cap-diode circuit

Circuit Design

- Consists of a multi-stage network of diodes (D2–D5) and capacitors (C4–C6).
- Stores and transfers charge during each switching cycle generated by the oscillator.
- Multiplies the input voltage to produce a higher DC output voltage at the output terminal.

